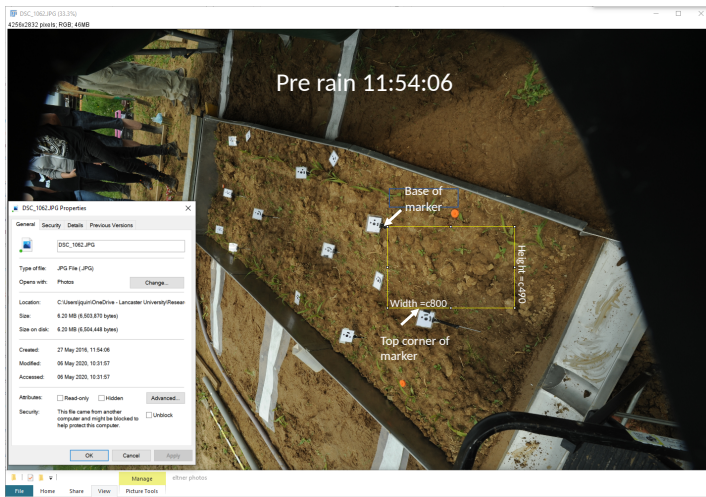


Detecting changes in covariance via random matrix theory

Rebecca Killick(r.killick@lancs.ac.uk)
JSM Aug 2023



- Motivation
- Inspiration
- F-matrix solution
- Simulations
- Application

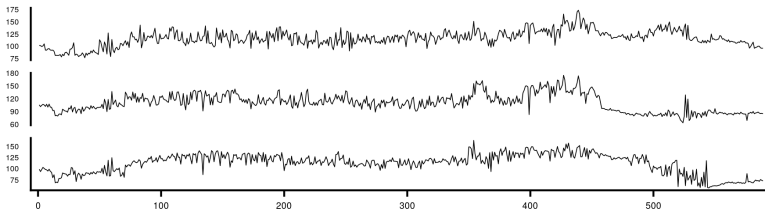


Motivation

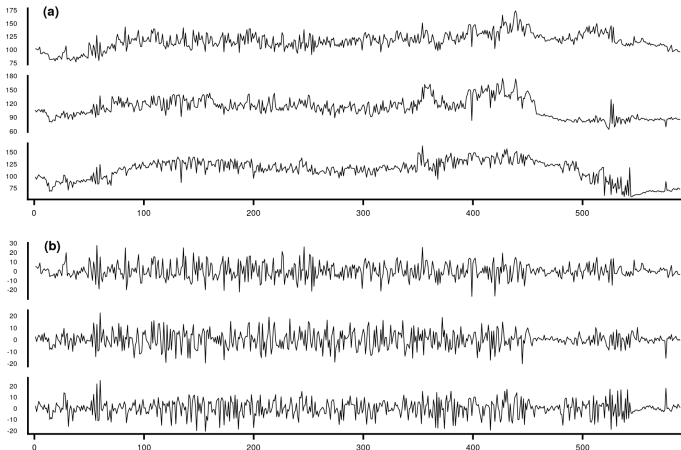


Motivation





Bayesian Trend Filter with Dynamic Shrinkage (Kowal et al. 2019), DSP R package



Detect multiple changes in covariance structure.

$$H_0 : \Sigma_0^* = \Sigma_{1,p} = \cdots = \Sigma_{n,p}$$

$$H_1 : \Sigma_1^* = \Sigma_{1,p} = \cdots = \Sigma_{\tau,p} \neq \Sigma_{\tau+1,p} = \cdots = \Sigma_{n,p} = \Sigma_2^*,$$

where τ is unknown.

Typically test statistics,

$$\psi_{\tau,1} \bar{\Sigma}(0, \tau) - \psi_{\tau,2} \bar{\Sigma}(\tau, n),$$

where $\{\psi_{\tau,1}\}_{\tau=\ell+1}^{n-\ell}$ and

$\{\psi_{\tau,2}\}_{\tau=\ell+1}^{n-\ell}$ are sequences of normalizing constants.

Large spectral norm $\implies$ changepoint.

Aue et al (2009), Galeano and Pena (2007), Wang et al (2017) . . .

All need estimate of Σ_0 for thresholds!

Univariate: use functions of $\frac{\sigma_1}{\sigma_2}$.

Advantages:

- More natural view, 1 under the null, deviations suggest changes
- No need to estimate the variance under the null
- Scale invariant



$$R(A, B) := B^{-1}A,$$

where $A, B \in \mathbb{R}^{p \times p}$.

$$T(A, B) = \sum_{j=1}^p (1 - \lambda_j(R(A, B)))^2 + \left(1 - \lambda_j^{-1}(R(A, B))\right)^2,$$

where $\lambda_j(R(A, B))$ is the j th largest eigenvalue of the matrix $R(A, B)$.

$$T(\Sigma_1, \Sigma_2) = \sum_{j=1}^p (1 - \lambda_j(R(\Sigma_1, \Sigma_2)))^2 + \left(1 - \lambda_j^{-1}(R(\Sigma_1, \Sigma_2))\right)^2,$$

- T is symmetric i.e. $T(\Sigma_1, \Sigma_2) = T(\Sigma_2, \Sigma_1)$;
- T is symmetric with respect to inversion of matrices i.e. $T(\Sigma_1, \Sigma_2) = T(\Sigma_1^{-1}, \Sigma_2^{-1})$;
- T does not depend on the underlying covariance.

If $\Sigma_1 = \Sigma_0 Z_1^T Z_1 \Sigma_0$ and $\Sigma_2 = \Sigma_0 Z_2^T Z_2 \Sigma_0$, then
 $T(\Sigma_1, \Sigma_2) = T(Z_1^T Z_1, Z_2^T Z_2)$.

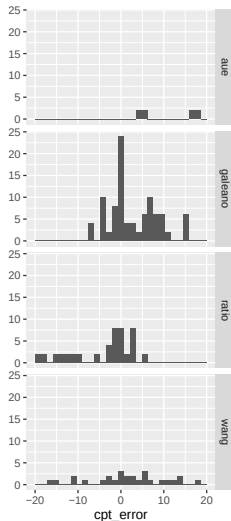
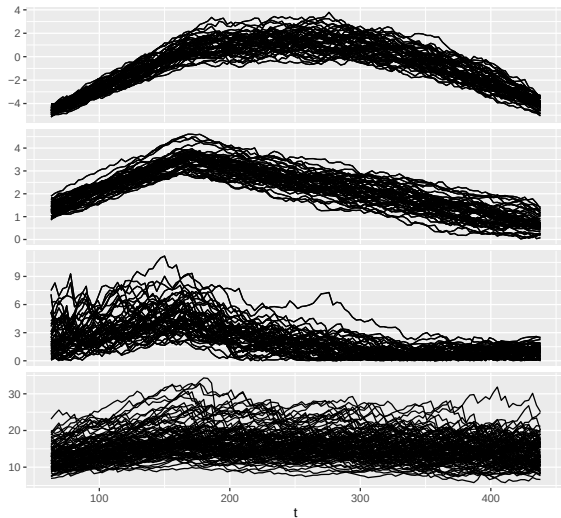
The sample sizes n_1, n_2 , and the dimension p grow to infinity such that

$$\gamma_{n_1} := \frac{p}{n_1} \rightarrow \gamma_1 \in (0, 1), \gamma_{n_2} := \frac{p}{n_2} \rightarrow \gamma_2 \in (0, 1) \text{ and } \gamma := (\gamma_1, \gamma_2).$$

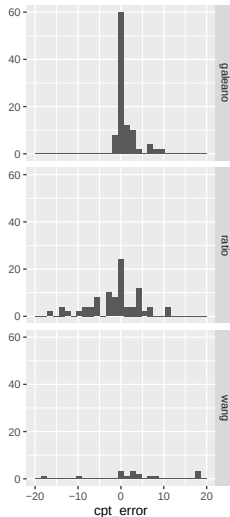
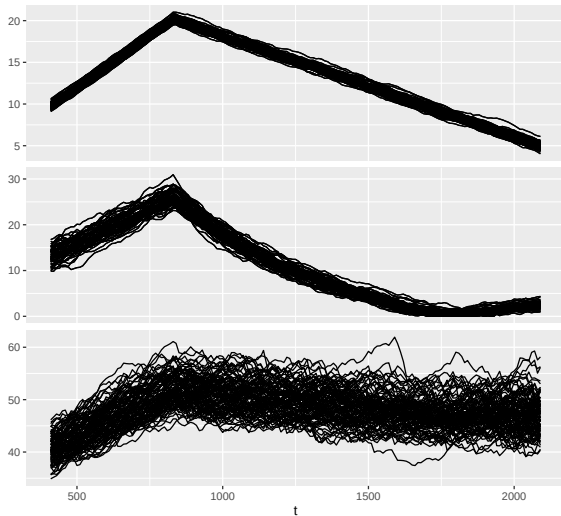
Then under some mild assumptions:

$$T \left(\frac{1}{n_1} X_{n_1,p}^T X_{n_1,p}, \frac{1}{n_2} X_{n_2,p}^T X_{n_2,p} \right) - p \int f^*(x) dF_\gamma(x) \rightarrow N(\mu(\gamma), \sigma^2(\gamma))$$

Single changepoint small



Single changepoint large



Multiple changepoints

p	n	FPR			
		Aue	Ratio	Wang	Galeano
3	500	0.36	0.24	0.63	0.00
	1000	0.48	0.28	0.77	0.02
	2000	0.54	0.31	0.85	0.09
	5000	0.59	0.31	0.90	0.23
10	500	0.25	0.16	0.27	0.00
	1000	0.32	0.13	0.43	0.04
	2000	0.34	0.10	0.53	0.09
	5000	0.37	0.09	0.62	0.21
30	2000		0.02	0.32	0.09
	5000		0.02	0.31	0.22
100	5000		0.00	0.45	0.26

Multiple changepoints

p	n	TPR			
		Aue	Ratio	Wang	Galeano
3	500	0.55	0.38	0.51	0.00
	1000	0.59	0.48	0.48	0.01
	2000	0.63	0.60	0.44	0.03
	5000	0.65	0.68	0.45	0.06
10	500	0.56	0.55	0.51	0.01
	1000	0.69	0.75	0.54	0.01
	2000	0.76	0.87	0.58	0.04
	5000	0.80	0.91	0.61	0.07
30	2000		0.98	0.45	0.05
	5000		0.99	0.56	0.10
100	5000		1.00	0.31	0.13

Multiple changepoints

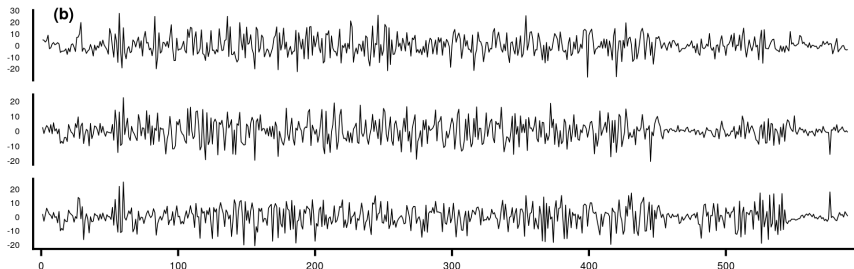
p	n	MAE			
		Aue	Ratio	Wang	Galeano
3	500	26.05	26.52	42.82	55.67
	1000	18.44	15.36	36.81	48.97
	2000	12.78	7.65	32.14	43.95
	5000	7.95	2.93	23.85	36.66
10	500	304.00	303.63	316.80	780.50
	1000	167.59	127.71	243.70	678.34
	2000	99.52	46.44	186.26	594.56
	5000	58.09	16.62	123.34	492.97
30	2000		143.43	1078.25	3079.91
	5000		51.37	525.14	2357.40
100	5000		199.51	7769.79	16281.43



Ratio: 64, 125, 184, 255, 340, 450, 527

Wang: 445 (huge eigenvalue affecting threshold)

Scientists: 64 dry - 76 very wet, increase 350, decrease 450



- New method based on ratio of covariance matrices
- Doesn't require estimation of Σ_0
- Good false positive control away from the boundaries
- Theoretical threshold works well for multiple changepoints

Future work

- Extend to mixed sampling rates and/or missingness